



Swing Trader · After-Fee Workflow

One question, asked the same way at every stage. **Is there a rule the model can trade that clears the fee on data it has never seen?** Nothing is accepted without passing that test, and nothing is reported that was not computed.

WORKFLOW CHEAT SHEET

seamusrobertmurphy/swing-trader
Crypto NO-GO · Equities live paper
State of 7 September 2026

1 Where data comes from EXCHANGE DIRECT

Price history is pulled from the official data, binance, vision archive, monthly per-symbol OHLCV zips, checksummed. It is **not** a live API, so a build is reproducible and offline.

SURVIVORSHIP COMPLETE The archive listing carries **612 PAIRS** ever listed against roughly 433 alive today, so delisted coins are included. A universe built from today's exchange listing silently deletes every coin that died, and every result on it is an upper bound.

PANELS ON DISK

FRAME	ROWS	COINS
5m	5.5M	11
1h	—	433
4h	3.9M	567
1d	520k	540
eq 1d	5.0M	2547

Stored as Parquet, never CSV. A round trip through CSV turns timestamps into strings and breaks the train and test split without raising anything.

2 Choosing a frame THE FEE DECIDES

A coarser bar trades fewer times, so it pays the toll less often. That is the whole argument, and it is why the 5-minute frame lost before the model was even consulted.



The indicator's own author says Supertrend bleeds below the 4-hour and holds up above it. The controls line ruled out 5-minute on the fee wall independently.

COARSER IS NOT FREE Fewer trades means fewer tolls but also less exposure per unit of return. The daily frame did not beat 4-hour; it traded the toll count against per-trade drift in a bear year.

13 Feature families 104 COLUMNS

PRICE, THE COIN'S OWN HISTORY
f_low, wad clock, f_high, intraday, f_low, indicators

SHAPE AND TREND
f_low, Supertrend, f_mid, adaptive, f_high, regime

CONTEXT BEYOND THE COIN
f_low, lead lag, f_high, f_d1, f_w1, f_high, crowding

ORDER FLOW
f_flow, order imbalance, f_mid, microstructure

Every feature is causal and scale-invariant, so a value uses only bars already closed and means the same on a \$90k coin as on a sub-cent one.

24 RAW LAGS MATCH ALL 104
Twenty-four lagged returns alone scored 24.247 against 24.213 for the engineered set. Both representations carry the same information, which is why a sequence model has no gap to close.

3 The four-gate screen ALL FOUR OR OUT

Run per bar, point in time, so a row is kept only if it would have passed on the day. A screen applied with today's knowledge is not a screen.

GATE	THRESHOLD
Liquidity	≥ 30M USDT 24h
Volatility	ATR 2.5–12%
Spread	≤ 0.05%
History	≥ 120 candles

The archives carry no order book, so spread is estimated by **Corwin-Schultz** from the high and low. The live screen uses the real spread.

RANK WIDE, TRADE NARROW The universe is scanned broadly, then a whitelist of coins that actually carried the out-of-sample edge is enforced at entry.

4 What a trade must clear THE FLOOR

Net move, never gross. A trade is refused unless what is left after the round trip clears the floor.

the whole fee argument in one line
net = move - round_trip - slippage
allow if net ≥ edge_floor

0.15 % achievable round trip
0.20 % standard round trip
1.5 % minimum net edge

0.075% per side with the exchange token discount and post-only maker entries, so no spread is paid. The live wrapper reads the account's **actual** rates each run and warns when the measured round trip exceeds the modelled one.

THIS IS THE WALL Crypto's real edge came out 5 to 12 basis points *smaller* than the 15 to 20 basis point round trip. No gate, label, frame or feature family closed that gap.

14 Cross-sectional RANK, DO NOT PREDICT

Two different questions. **Will this coin rise?** failed everywhere. **Which of these coins is strongest right now?** did not.



TWO STATEMENTS, KEPT APART The ranking signal is real and was never disproved. The BTC-gated way of trading it was killed at 27% of folds against the 60% bar. Those are different claims.

23 Reading the fold chart SPREAD BEATS MEAN

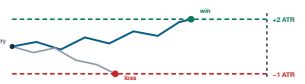


Bars near the dashed line generalise. Bars well above it memorised. Fold 1 is worse than simply guessing the average.

NEVER READ THE MEAN ALONE One aggregate hides which periods worked. Four folds ranging 22.3 to 27.0 is a different object from a steady 24.9, and only the folds tell you which you have.

5 The label TRIPLE BARRIER

What counts as a win is a decision, not a fact. Three barriers settle it: a profit target, a stop, and a clock. Whichever is touched first is the answer.



Barriers scale with ATR, so a calm coin and a violent one are judged on the same terms rather than the same percentage.

A LABEL CAN LOSE BEFORE THE MODEL SPEAKS The 4-hour default hits its target on **0.269** of bars while breakeven needs **0.333**. Negative expectancy is built in, so no model on it can win. Check the base rate against breakeven first.

6 Splitting time NEVER AT RANDOM

A random split puts next Tuesday in training and scores the model on last Monday. Market bars are autocorrelated, so that leaks the future backwards and flatters everything.



Embargo is a gap at the cut, as wide as the label horizon, so no label straddles it. Without it the last training rows carry outcomes that live in the test window.

15 Model scorecard FULL VS CROSS VALIDATED

MODEL	FULL		CROSS VAL.		RATIO
	MAE	RMSE	MAE	RMSE	
HistGBM	.2181	.2619	.3861	.4818	1.840
LightGBM	.2387	.2806	.4036	.4826	1.720
RF	.4445	.4526	.4761	.4840	1.069
LogReg	.4718	.4855	.4463	.4909	1.011

4-hour frame, 7 Sep 2026. RMSE on predicted probabilities is the square root of the Brier score.

THE WINNER FAILS THE RULE
Selection takes the best cross-validated RMSE and never consults the ratio, so it crowned HistGBM at **1.840** while the bar rejects anything over 1.1. The two stable models sat 2 to 5% behind on error.

19 Symptom and fix ALL SEEN HERE

SYMPTOM	FIX
Page renders, numbers wrong	Compare against the previous render. Exit zero proves nothing.
Ignored file still on GitHub	git rm --cached; a rule alone never untracks.
Render stops, no result	One slow cell aborts the run. Set error: true.
Tables vanish in Word	Pandoc drops raw markdown blocks. Render through Quarto.

7 The four numbers READ TOGETHER

RMSE average miss, big misses squared first so one disaster moves it
MAE average miss, every miss counted once
MAPE average miss as a share of truth; needs a target that is never zero
ratio CV RMSE ÷ train RMSE

When RMSE far exceeds MAE the model is **occasionally very wrong** rather than steadily slightly wrong. That distinction changes what you fix.

THE RATIO WAS UPSIDE DOWN UNTIL 7 SEP 2026 It divided train by CV, so an overfit model scored 0.91 where the rule expects 1.10, and anyone applying "reject above 1.1" passed exactly what it exists to catch.

≤ 1.1 stable, keep
> 1.1 memorised, reject

8 What tuning showed LESS, NOT MORE

Nine settings, three walk-forward folds, 4-hour frame, predicting bars until the trend flips.

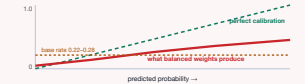


Training error nearly halves while held-out error gets worse. Every one of the nine settings is rejected on the 1.1 bar, and the winner was the weakest model in the grid.

PERIOD BEATS CONFIGURATION, 40 TO 1 Fold spread inside one setting is 3.83 bars. The largest gap between any two settings is 0.094.

16 The calibration hole NOTHING BUILT YET

Every model trains with **balanced class weights**, which deliberately pushes predicted probabilities toward 0.5 and away from the true base rate. The headline metric is Brier, which measures exactly that fit.



This curve does not exist in the repository. There is no reliability plot, no Platt scaling, no isotonic step. One comment notes the Brier relationship and that is all.

WORSE THAN A CONSTANT Always predicting the base rate scores **0.4155** on daily; the best model scores **0.4662**. Twelve per cent worse. It does not mean no skill, it means the probabilities are wrong even where the ranking is not.

9 Passing the bar FOLDS, NOT TOTALS

A total can be carried by one lucky stretch. History is cut into half-year folds and the edge must hold in **60 per cent** of them, on two questions at once.

TEST	ASKS
Absolute	does the picked cohort make money after fees
Selection	does it beat the market in the same fold

ABSOLUTE ALONE IS A TRAP Markets drift up and a survivor-only panel drifts harder. Beta plus survivorship can fake an absolute positive on their own, so the top-minus-market spread is the real bar.

SURVIVES both bars, kept
PARTIAL closer, not clear
KILLED falsified

10 Where it stands HONEST SCOREBOARD

4h ungated -0.382
+ BTC gate -0.117
+ funding -0.067
breakeven 0.000

Per-trade return after fees, out of sample. A year of work moved the line from -0.38 to -0.05, and none of it crossed zero.

THEY LOSE BEFORE FEES Edge diagnostics, 7 Sep: daily -1.150% and 4-hour -0.182% per trade pre-cost. That reframes a year of calling this a fee-wall problem.

THE ONE LIVE EDGE

12-1 equity momentum. Buy what rose over the last year, skipping the most recent month. Spread +1.082%/month over 116 months, 1,241, clearing 85% and 80% of folds. Paper book live since 18 Aug.

17 The live book PAPER, REAL PRICES



Fourteen sessions. The plan expects swings of about 8.2% over that span, so a 3.1% dip is 0.4 of an ordinary move rather than a signal.

50 names @ 1.8%
3/6 clean cycles
7.3 bp per fill mid-session

THIS IS FAKE MONEY A paper account at real prices. LIVE_TRADING is off and every order path aborts unless it is the exact string true.

20 Before deploying ALL EIGHT

A check is finished when all of these hold. Not a check.

- Base rate beats breakeven.** Check the label's own expectancy before any model is fitted.
- Split is forward-chained** with an embargo as wide as the label horizon.
- Blind window scored once.** A threshold chosen by looking at it is no longer blind.
- Both fold bars cleared,** absolute and selection, at 60 per cent or better.
- Overfit ratio at or under 1.1,** and selection consulted it.

11 Traps that cost days ALL SEEN HERE

- Rendering is not checking.** A page can build, exit zero, and report 0 of 6 cycles instead of 3. Compare against the previous render.
- Gitignore does not untrack.** A rule only stops files git does not already track. git ls-files -i -c --exclude-standard lists what needs removing.
- Paths built from pieces.** A search for "outputs/" misses join(x, "outputs", y). That wrote 51MB to a stray folder, twice, silently.
- Balanced weights break Brier.** Class balancing pushes probabilities off the base rate while Brier measures exactly that fit. Every model scored worse than a constant.
- A timeout hides everything after it.** One slow cell aborted a run and concealed 20 cells behind it.

12 Running it SCRIPTS, NOT NOTEBOOKS

```
# weekly, intraday only
alpaca_data.py download
alpaca_trade.py rebalance --execute
alpaca_trade.py check
alpaca_execution_report.py
```

NEVER OUTSIDE THE SESSION
Orders queued into the opening auction cost **42.6BP** per fill against **7.3BP** mid-session. Same book, eleven times the cost, from the clock alone.

HARD RULES IN CODE

5% position cap
3% daily circuit
10% cash floor

Never short, never margin, never options, never average down. Enforced in code, not assumed from the venue. LIVE_TRADING must be the exact string true or every order path aborts.

18 What killed what CLOSED ARTEFACTS

CANDIDATE	KILLED BY
1h direction	AUC 0.50, coin-flip floor
Entry sharpening	29% of folds, regime artefact
4h	
MST + breadth gate	27% of folds vs 60% bar gate
Funding gate	-0.067% at achievable cost
Daily frame	label degenerate, base rate 0.068
Microstructure	no selection edge
5m scalp	fee wall
Sequence models	0.25% gap, priced not worth it

A KILL IS A RESULT Each one is written up with its numbers and kept. The harness was validated by killing 6-1 momentum and low-vol exactly as the literature predicts, which is why its verdicts are trusted.